# Kyle Onghai

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## Education

**University of California, Los Angeles** **Los Angeles, CA**
*B.S. Mathematics, Specialization in Computing* *September 2020 – Expected 2024*

- Honors College, Dean's List, Alumni Scholar's Club Member
- Relevant Coursework: Honors Algebra, Honors Real Analysis, Honors Linear Algebra, Honors Numerical Analysis, Probability, Differential Geometry, Complex Analysis, Nonlinear Differential Equations, Discrete Mathematics, Physics, Introduction to Programming, Python and Applications, Web Programming
- Cumulative GPA: 3.87, Major GPA: 3.86

**Earl L. Vandermeulen High School** **Port Jefferson, NY**
*High School Diploma* *September 2016 – June 2020*

- Graduated Salutatorian
- Received the New York Regents Scholarship
- National AP Scholar, National Honor Society Member

## Activites, Awards, and Leadership

**University of California, Los Angeles (UCLA)** **Los Angeles, CA**

- Top 500 on 81st William Lowell Putnam Mathematical Competition (2020)
- Member of the UCLA ICPC Club that sends teams to compete at the ACM ICPC

## Research/Work Experience

**Massachusetts Institute of Technology** **Cambridge, MA**
*Summer Geometry Initiative 2023* *July 2023*

**Centre de Researches Mathematiques** **Montreal, QC**
*12th Montreal Industrial Problem Solving Workshop* *August 2022*

- Built a prototype for the International Air Transport Association to map air turbulence using real-time eddy dissipation rate data.
- Allows pilots to avoid patches of high air turbulence in order to reduce wasteful fuel consumption and number of injuries.

**Moravian University** **Bethlehem, PA**
*REU: Research Challenges of Computational Methods in Discrete Mathematics* *June 2022 – August 2022*

- Established the theory of distribution graphs in the study of chip-firing games on simple graphs.
- Described the contractions of 2-edge-connected graphs with maximum matching number at most 4.

**Collaborative Mathematical Research Program** **Online**
*Undergaduate Researcher* *June 2021 – August 2021*

- Showed that there are infinitely many $k$-Diophantine $m$-tuples over a finite field with a sufficiently large, odd prime cardinality.
- Proved a closed-form formula for the number of 3-Diophantine triples in a finite field with an odd prime cardinality.
- Developed an asymptotic formula for the number of $k$-Diophantine $m$-tuples in a finite field with an odd prime cardinality.

**Stony Brook University** **Stony Brook, NY**
*Simons Summer Research Fellow* *June 2019 – August 2019*

- Added photoacoustic tomography functionality to Orthopaedic Bioengineering Lab's proprietary quantitative ultrasound machine.
- Devised analytic method in MATLAB to reconstruct images of phantoms using Fourier transforms.

## Skills

**Software**: Python, LaTeX, SAGE, C/C++, MATLAB, Lua, BASH, Linux, Rust
**Hardware**: Raspberry Pi, Arduino
**Languages**: English (Native), Mandarin (Intermediate), Latin

## Community Involvement

**Stony Brook University Medical Center** **Stony Brook, NY**
*November 2018 – February 2020* *Patient Education Volunteer*

- Taught patients how to access and use the Patient Portal, and received the Presidential Award for Volunteer Service.